

The Hidden Map of Language

A Journey into Word Senses, Relations, and WordNet

Based on the lectures of Dr. Chetana Gavankar

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1 Introduction: The Chaos of Meaning

Imagine you walk into a room and say, "**I'm going to the bank.**"

A friend might picture you walking towards a river to fish. Another might imagine you depositing a check. This ambiguity is the heart of the problem we're solving today. Computers, unlike humans, don't have the shared context of "going to the bank" usually implying finance unless we're carrying a fishing rod.

Today, we will unravel:

- **Word Senses:** The atomic units of meaning.
- **Relations:** How words are connected (like a family tree).
- **WordNet:** The grand map of these connections.
- **Word Sense Disambiguation (WSD):** The detective work of figuring out which "bank" we mean.

2 Part 1: The Ambiguity of Language

The Tale of Two Banks

Imagine our protagonist, the word "**Bank**". By day, Bank is a serious, suit-wearing financial institution holding your investments. But on the weekends, Bank relaxes by the river, becoming a muddy slope where you can sit and fish.

These are not two different people; they are the same spelling, the same pronunciation, but two completely different *lives*. In NLP, we call these different lives **Senses**.

2.1 The Anatomy of a Word

Before we dive deeper, let's define our terms. Words aren't just strings of letters; they have structure.

Key Concept: Lexeme, Lemma, and Sense

- **Lexeme:** An entry in the dictionary (lexicon) that pairs a form (spelling/sound) with a meaning.
- **Lemma:** The "citation form" or the dictionary headword. Think of it as the representative of the word family.
 - *Example:* "Carpet" is the lemma for "carpets". "Sing" is the lemma for "singing, sang, sung".
- **Sense:** A discrete representation of one aspect of the meaning of a word.

2.2 The Web of Relationships

Words don't live in isolation. They have relationships with each other, much like people in a community.

2.2.1 1. Homonymy: The Identical Strangers

Analogy Time: The Doppelgangers

Homonyms are like two strangers who happen to look exactly alike (same spelling/sound) but are completely unrelated. They don't share DNA; they just share a face.

Example:

- **Bat** (the wooden stick used in sports)
- **Bat** (the flying mammal)

These words share a form (phonological or orthographic) but have unrelated, distinct meanings. This causes huge headaches for:

- *Text-to-Speech*: "Bass" (fish) vs. "Bass" (instrument) are pronounced differently!
- *Information Retrieval*: Searching for "bat care" might give you tips on oiling a cricket bat or feeding a fruit bat.

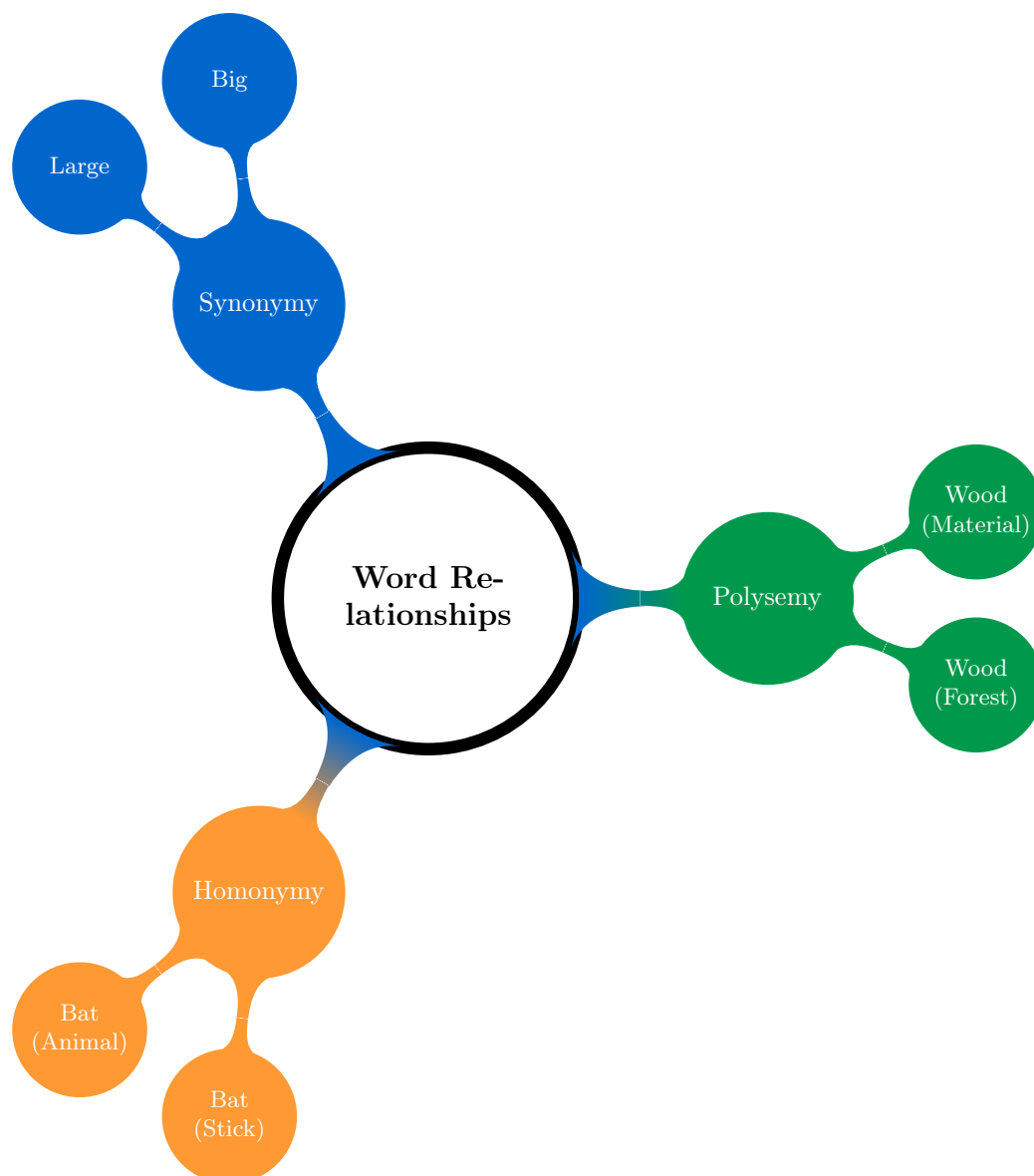
2.2.2 2. Polysemy: The Versatile Actor

Unlike Homonyms, Polysemous words are related. They are like an actor playing different roles that share a core essence.

Example: "Bank" (financial institution) and "Bank" (building).

- "The bank raised interest rates." (Institution)
- "The bank is on Main Street." (Building)

Are these different senses? Yes. Are they related? Absolutely.



2.2.3 3. Synonymy: The Twins

Synonyms are words with the same meaning in some or all contexts.

- *Couch / Sofa*
- *Big / Large*

But wait! Are they perfect twins?

"Miss Nelson became a kind of **big** sister to Benjamin."

"Miss Nelson became a kind of **large** sister to Benjamin." (?)

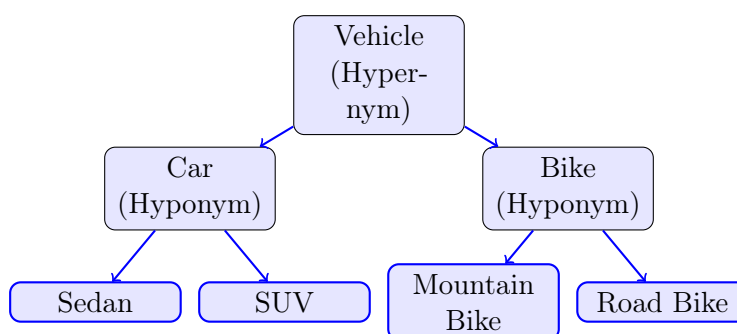
"Big" implies age or maturity here, while "large" implies size. So, true synonymy is rare. It's a relationship between *senses*, not just words.

2.2.4 4. Hyponymy Hypernymy: The Family Tree

This is the "is-a" relationship.

- **Hyponym:** The specific child (Subclass). E.g., *Mango* is a hyponym of *Fruit*.

- **Hypernym:** The general parent (Superordinate). E.g., *Vehicle* is a hypernym of *Car*.



3 Part 2: The Map of Meaning - WordNet

The GPS of Language

If words are cities and villages, how do we find the roads connecting them? We need a map.

Enter **WordNet**. Created at Princeton, WordNet is not just a dictionary. It's a massive, hierarchically organized lexical database. It's the Google Maps of English.

3.1 The Building Block: The Synset

In a normal dictionary, the unit is a word. In WordNet, the unit is a **Synset** (Synonym Set).

Key Concept: What is a Synset?

A synset is a group of synonyms that represent the *same concept*.

- **Example:** {chump, fool, gull, mark, patsy}
- **Gloss:** A person who is gullible and easy to take advantage of.

WordNet doesn't just link words; it links *concepts*.

3.2 Navigating the Hierarchy

WordNet is organized like a massive tree (or rather, a graph). Nouns, Verbs, Adjectives, and Adverbs live in their own hierarchies.

3.2.1 Noun Relations

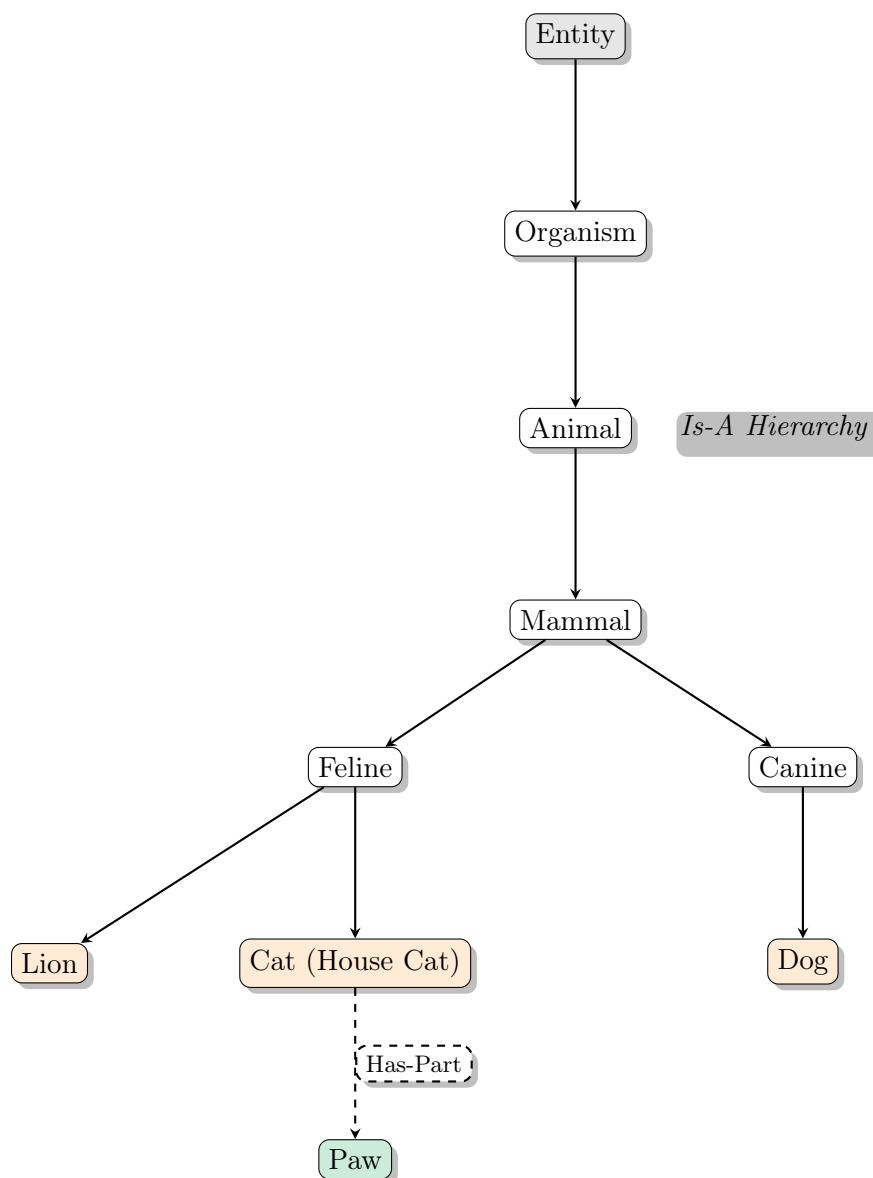
- **Hypernym:** The parent (Vehicle is a hypernym of Car).
- **Hyponym:** The child (Car is a hyponym of Vehicle).
- **Meronym:** The "part-of" relationship.
 - *Part Meronym:* Wheel is a part of a Car.
 - *Member Meronym:* Faculty is a member of a Committee.
- **Antonym:** Opposites (Dark vs. Light).

3.2.2 Verb Relations

- **Hypernym:** To Move is a hypernym of To Walk.
- **Troponym:** The specific "manner" of doing something. (To March is a troponym of To Walk).
- **Entailment:** If you Snore, it entails you Sleep.

3.3 Visualizing the Map

Let's see what a slice of this map looks like.



Analogy Time: The Depth of Specificity

Notice how "Entity" is vague, but as you go down, it gets specific.

- **Root:** Entity (100% abstraction)
- **Leaf:** Cat (Concrete reality)

The deeper you go, the more information you have.

4 Part 3: The Detective Work - Word Sense Disambiguation (WSD)

The Case of the Ambiguous Bass

Imagine you are Sherlock Holmes. You find a note: "He caught a huge bass."
Is it a fish? Is it an instrument? Without context, it's impossible. But what if the note says: "He caught a huge bass **in the river**."?
Aha! "River" is the clue. This detective work is what we call **Word Sense Disambiguation (WSD)**.

4.1 The WSD Task

Given:

1. A word in context (e.g., "The **bank** can guarantee deposits...").
2. A fixed inventory of potential senses (e.g., Sense 1: Finance, Sense 2: River).

The Goal: Decide which sense applies.

Key Concept: Two Types of Investigations

- **Lexical Sample Task:** You focus on a specific suspect. (Disambiguate "bank" and "plant" only).
- **All-Words Task:** You must solve the mystery for *every single word* in the text. Harder!

4.2 The Baseline: Just Guess the Popular One

Before we use fancy algorithms, let's try a simple trick: Just pick the sense that appears most often.

- **Most Frequent Sense (MFS):** In WordNet, senses are ordered by frequency. Sense 1 is the most common.
- **Surprise:** This is a tough baseline to beat! If 80% of "banks" are financial, guessing "financial" is 80% accurate.

4.3 Supervised Learning: Training the Detective

We can train a machine learning classifier (like SVM or Naive Bayes) using labeled examples. We extract **features** from the context.

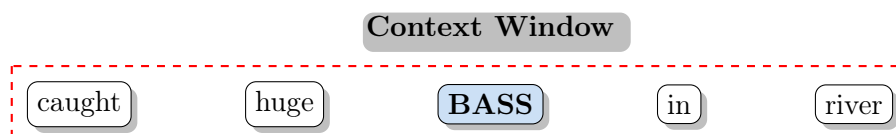
Analogy Time: Clues in the Window

Imagine looking through a window around the target word. The words inside are your clues.

4.3.1 Feature Types

- **Collocational Features:** Specific words at specific positions. (e.g., "word to the left is 'river'").

- **Bag-of-Words Features:** Just counts of words in the window, ignoring position. (e.g., "fish" appears 2 times).



4.4 The Lesk Algorithm: Knowledge is Power

What if we don't have labeled training data? We can use a dictionary (like WordNet) directly.

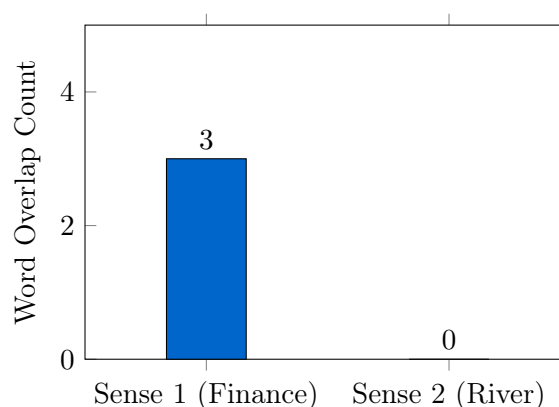
Key Concept: The Lesk Intuition

The correct sense's definition (gloss) probably shares words with the context.

Example:

- **Context:** "The *bank* can guarantee **deposits**..."
- **Sense 1 (Finance):** "...financial institution that accepts **deposits**..."
- **Sense 2 (River):** "...sloping land beside a body of water..."

Overlap: "deposits". **Winner:** Sense 1.



4.5 Wikipedia as a Resource

We can also use Wikipedia! Wikipedia pages are like senses. Links (e.g., [[Bar (Law)]] vs. [[Bar (Music)]) act as labeled data. This helps us build massive datasets for training.

5 Conclusion

We have journeyed from the atomic ambiguity of a single word to the vast network of WordNet, and finally to the algorithms that help machines understand meaning.

- Words are ambiguous.
- WordNet maps their relations.
- WSD algorithms (Supervised Lesk) solve the ambiguity using context.